

Alex Lewandowski

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EDUCATION

University of Waterloo — B.A.Sc., Mechatronics Engineering
President's Scholarship of Distinction

Sep 2026 – Jun 2031

EXPERIENCE

Head Programmer, Designer & Strategist

Jun 2022 – Jun 2026

VEX Robotics Team 5225A, The PiLons

- Won the **Excellence Award**, the top team in all of Ontario, at 2025 Provincials to advance to VEX Worlds, placing **4th of 10,000+ teams** worldwide in Autonomous Skills and earning the **Design Award** for an organized, professional engineering design process and notebook
- Won the **Think Award** at the **2024 and 2026** VEX Worlds, recognizing clean, well-documented autonomous code and a clearly explained programming strategy
- Led iterative mechanical design in Fusion 360, redesigning the robot between events to adapt to new strategy
- Engineered a power take-off (PTO) system that repurposed the X-drive motors to power the hanging mechanism, **an approach no other X-drive team had implemented that season**
- Built an odometry system fusing gyroscope, encoder wheel, and distance-sensor data to track robot position to **± 0.02 m**
- Applied kinematics and inverse kinematics to control robotic arm positioning for precise game-element manipulation
- Designed closed-loop PID and feedforward velocity controllers to drive real-time autonomous motion algorithms
- Developed state machines to coordinate subsystem behavior across autonomous and driver-controlled phases

Team Captain, Head of Programming

Sep 2019 – May 2022

First Lego League Robotics Team

- Programmed autonomous routes (line-following, gyro turns) in Python; planned the route that scored **710 points, 30 above the prior world record**
- Led team to Championship at Western Edge (California, 2022) and first-place provincial finishes, including the **Engineering & Excellence Award (2020)**

PROJECTS

World Robot Olympiad Autonomous Robot | *ESP32, Teensy 4.1, Sensor Integration, CAD* Oct 2025 – May 2026

- Qualified for the **California Invitational Open** after Nationals, where the robot was the only one to use pneumatics and custom-designed electronics
- Split the robot's compute between an ESP32 for sensing and localization and a Teensy 4.1 for motor control, integrating a **9-axis IMU**, dual color sensors, and a camera module for field-element detection
- Designed and 3D printed a compact chassis in CAD to house the electronics, motors, and sensors within the competition's size limit

Differential Swerve Drivetrain | *Fusion 360, Mechanical Design, Load Testing*

Sep 2025 – Dec 2025

- Designed each module as a mechanical differential, where summing the two motor speeds drives the wheel and their difference steers it
- Iterated the chassis after load testing exposed C-channel bending, reinforcing it with printed triangle braces and a custom steel-ball thrust bearing

Dexterous Robotic Hand | *3D Printing, Mechanism Design, Gear Design*

Nov 2025 – Jan 2026

- Designed and 3D printed a **five-fingered** robotic hand with a rotating wrist and elbow, iterating the finger joint geometry from circular string holes to full-length slits and adding a gear-driven bearing assembly for wrist rotation

SKILLS

Robotics & Controls: PID, feedforward, odometry, sensor fusion, kinematics, state machines

CAD & Mechanical Design: Fusion 360, SolidWorks (in progress), 3D printing

Manufacturing & Prototyping: Iterative prototyping, load testing, mechanical fastening design

Embedded Systems: PROS, OkapiLib, ESP32, Teensy 4.1, I2C

Languages: Python, C++, Java

Machine Learning & CV: OpenCV, NEAT (neuroevolution)

LEADERSHIP & ACTIVITIES

Competitive Sailing Skipper — 420 Race Team; 6 years of sailing, 13+ regattas, 10th at Cork International Regatta; volunteered helping younger sailors with boat handling and racing fundamentals